LABSHEET=02 PRIYANSHU SHARMA CU240250980 SECTION=(B) (47) Mastery in Data Analytics and Visualizations and Career Advancement (Technical)

```
# Contrast Enhancement and Histogram-Based Image Processing
# Using Python and OpenCV
# ============================================================

import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
# 1. LOAD IMAGE
# ============================================================

image_path ="/content/pexels-friededia-30649280.jpg"  # Put your image in the same folder

image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image not found!")
    print("Make sure 'input.jpg' is in the same folder as this Python file.")
    exit()

print("Image loaded successfully!")
print("PRIYANSHU SHARMA CU240250980")
```

```
Image loaded successfully!
PRIYANSHU SHARMA CU240250980
```

```
# 2. CONVERT IMAGE TO GRAYSCALE
# ============================================================

gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# 3. CONTRAST STRETCHING
# ============================================================

min_pixel = np.min(gray)
max_pixel = np.max(gray)

if max_pixel != min_pixel:
    contrast_stretched = cv2.normalize(
        gray,
        None,
        0,
        255,
        cv2.NORM_MINMAX
    )
else:
    contrast_stretched = gray.copy()
```

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# 4. HISTOGRAM EQUALIZATION
# ============================================================

hist_equalized = cv2.equalizeHist(gray)
```

```
# 5. CLAHE
# ============================================================

clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray)
```

```
# 6. DISPLAY ORIGINAL AND ENHANCED IMAGES
# ============================================================

plt.figure(figsize=(14, 8))

plt.subplot(2, 2, 1)
plt.imshow(gray, cmap="gray")
```

```
plt.title("Original Grayscale Image")
plt.axis("off")

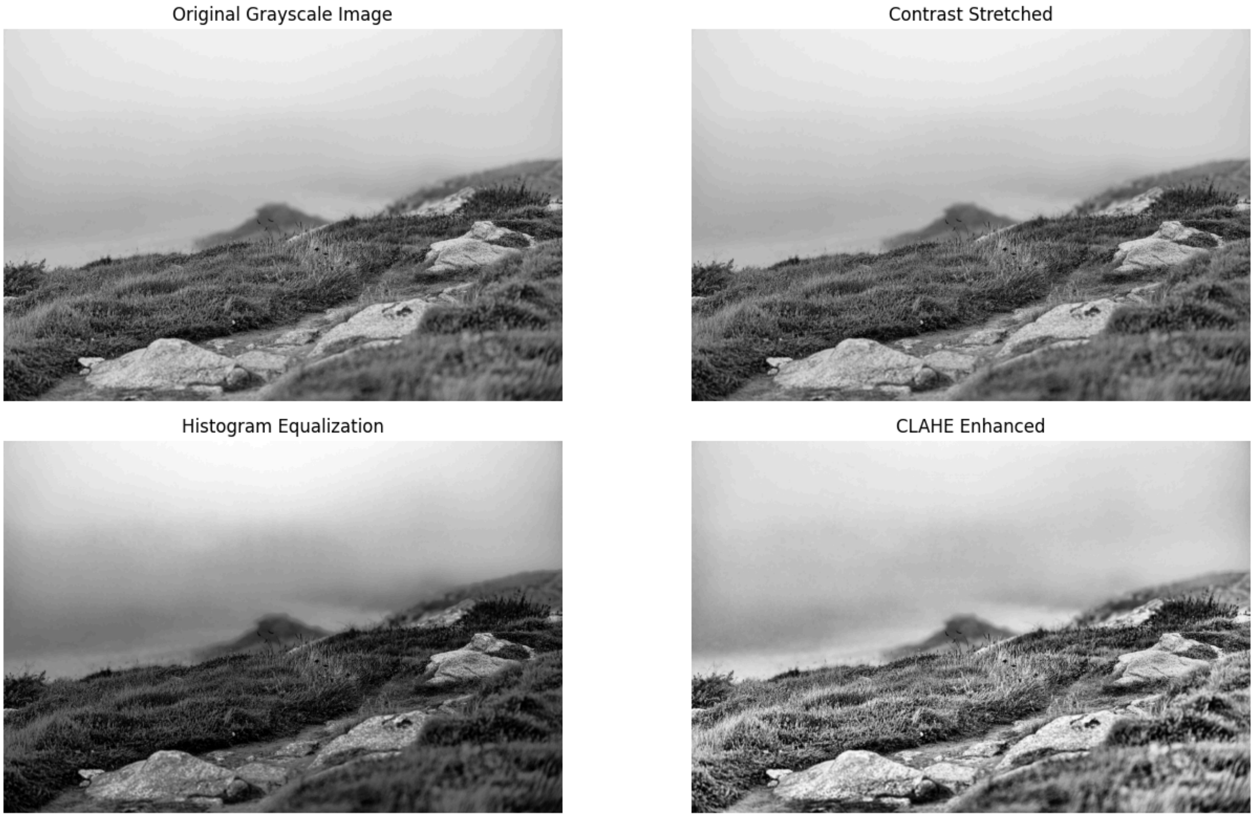
plt.subplot(2, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE Enhanced")
plt.axis("off")

plt.tight_layout()
plt.show()


# ============
```

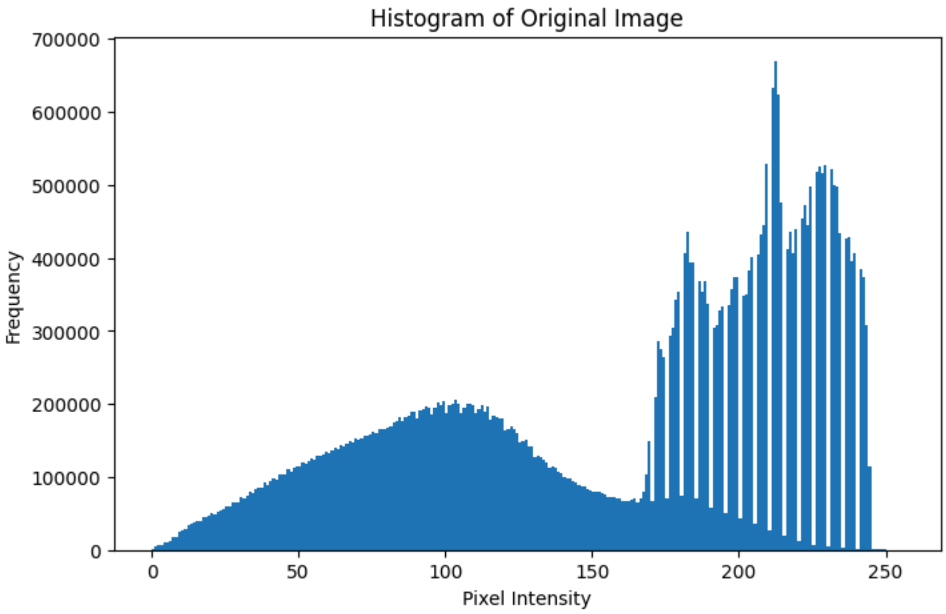


```
# 7. HISTOGRAM OF ORIGINAL IMAGE
# ============================================================

plt.figure(figsize=(8, 5))

plt.hist(
    gray.ravel(),
    bins=256,
    range=[0, 256]
)

plt.title("Histogram of Original Image")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
```




```
# 8. COMPARE ALL HISTOGRAMS
# ============================================================

plt.figure(figsize=(12, 8))

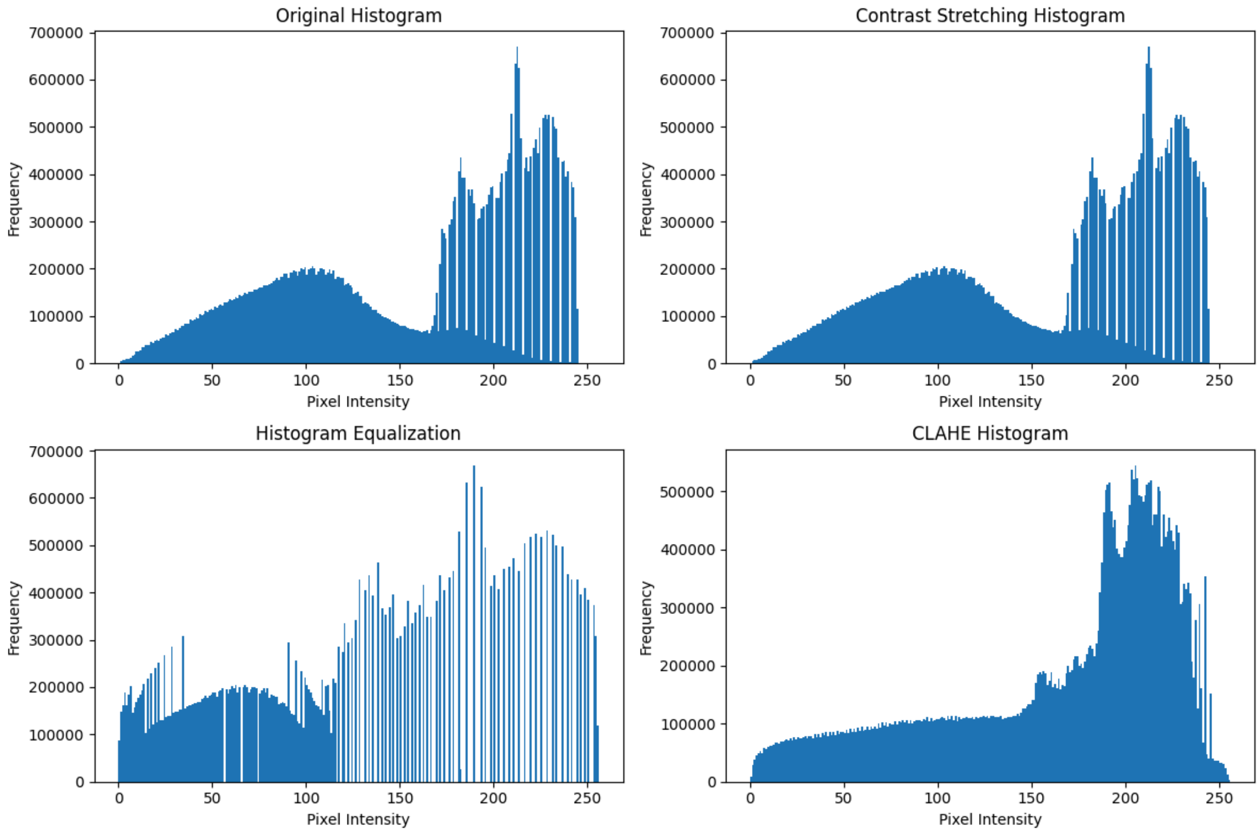
plt.subplot(2, 2, 1)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 2)
plt.hist(
    contrast_stretched.ravel(),
    bins=256,
    range=[0, 256]
)
plt.title("Contrast Stretching Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 3)
plt.hist(
    hist_equalized.ravel(),
    bins=256,
    range=[0, 256]
)
plt.title("Histogram Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 4)
plt.hist(
    clahe_image.ravel(),
    bins=256,
    range=[0, 256]
)
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
```

```
plt.tight_layout()
plt.show()
```




```
# 9. SAVE ENHANCED IMAGES
# ============================================================

cv2.imwrite("original_gray.jpg", gray)
cv2.imwrite("contrast_stretched.jpg", contrast_stretched)
cv2.imwrite("histogram_equalized.jpg", hist_equalized)
cv2.imwrite("clahe_enhanced.jpg", clahe_image)

print("\nAll enhanced images have been saved successfully!")
```

```
All enhanced images have been saved successfully!
```

```
# 10. BASIC OBSERVATION
# ============================================================

print("\n========== OBSERVATION ==========")
print("1. Original image shows the original contrast.")
print("2. Contrast stretching expands the intensity range.")
print("3. Histogram Equalization improves global contrast.")
print("4. CLAHE improves local contrast and works well")
print("   for images with uneven illumination.")
print("=================================")
```

```
========== OBSERVATION ==========
1. Original image shows the original contrast.
2. Contrast stretching expands the intensity range.
3. Histogram Equalization improves global contrast.
4. CLAHE improves local contrast and works well
   for images with uneven illumination.
=================================
```